

AGILE SOFTWARE ENGINEERING (CS303)

Scrum Guide Summary

Learning Objectives

Understand Scrum, its roles, events, artifacts, values, pillars, workflow and applications in Agile software development.

Introduction to Scrum

Scrum is a lightweight Agile framework used to manage and develop complex products through iterative and incremental delivery.

What is Scrum?

Scrum helps teams work together, adapt to change, and continuously improve product quality through short development cycles called Sprints.

Three Pillars of Scrum

1. Transparency 2. Inspection 3. Adaptation

Five Scrum Values

Commitment, Focus, Openness, Respect, and Courage are the core values that guide Scrum teams.

Scrum Roles

1. Product Owner – Represents stakeholders and manages Product Backlog.
2. Scrum Master – Facilitates Scrum practices and removes impediments.
3. Development Team – Builds and delivers the product increment.

Scrum Events

Sprint → Sprint Planning → Daily Scrum → Sprint Review → Sprint Retrospective

Sprint

A Sprint is a fixed time-boxed iteration, usually lasting 1–4 weeks, during which a potentially shippable product increment is developed.

Daily Scrum

A 15-minute daily meeting where team members discuss progress, upcoming work, and blockers.

Sprint Review

The completed work is demonstrated to stakeholders and feedback is collected.

Sprint Retrospective

The Scrum Team reflects on its performance and identifies improvements for future sprints.

Scrum Artifacts

1. Product Backlog
2. Sprint Backlog
3. Increment

Scrum Workflow

Product Backlog → Sprint Planning → Sprint Backlog → Development → Increment → Review → Retrospective → Next Sprint

Advantages of Scrum

Improved collaboration, faster delivery, better quality, flexibility, and increased customer satisfaction.

Limitations of Scrum

Requires active team participation, experienced members, and continuous stakeholder involvement.

Real World Applications

Scrum is widely used by software companies such as Microsoft, Amazon, Adobe, Spotify, and Google.

Exam Important Questions

1. Define Scrum Framework.
2. Explain Scrum Roles.
3. Discuss Scrum Events.
4. Explain Scrum Artifacts.
5. Describe the Scrum Workflow with a neat diagram.

Summary

Scrum is the most popular Agile framework. It provides a structured approach for delivering products through collaboration, transparency, and continuous improvement.